4. Vector Spaces

4.1 Vector Spaces and Subspaces

Definition: Vector Space

A vector space is a nonempty set V of objects, called vectors, on which are defined two operations, called addition and multiplication by scalars (real numbers), subject to the ten axioms (or rules) listed below. The axioms must hold for all vectors $\vec{u}$, $\vec{v}$ and $\vec{w}$ in V and for all scalars c and d.

1. The sum of $\vec{u}$ and $\vec{v}$, denoted by $\vec{u}+\vec{v}$, is in V.
2. $\vec{u}+\vec{v}=\vec{v}+\vec{u}$
3. $(\vec{u}+\vec{v})+\vec{w}=\vec{u}+(\vec{v}+\vec{w})$
4. There exists a vector $\vec{0}$ in V such that $\vec{u}+\vec{0}=\vec{u}$
5. For each $\vec{u}$ in V, there is a vector $-\vec{u}$ in V such that $\vec{u}+(-\vec{u})=\vec{0}$
6. The scalar multiple of $\vec{u}$ by c, denoted by $c\vec{u}$, is in V
7. $c(\vec{u}+\vec{v})=c\vec{u}+c\vec{v}$
8. $(c+d)\vec{u}=c\vec{u}+d\vec{u}$
9. $c(d\vec{u})=(cd)\vec{u}$
10. $1\cdot\vec{u}=\vec{u}$

Corollaries of Vector Space Axioms

For any $\vec{u}$ in V and scalar c:

$$0\vec{u}=\vec{0}$$
$$c\vec{0}=\vec{0}$$
$$-\vec{u}=(-1)\vec{u}$$

Definition: Subspace

A subspace of a vector space V is a subset H of V that has three properties:
a. The zero vector of V is in H.
b. H is closed under vector addition. That is, for each $\vec{u}$ and $\vec{v}$ in H, the sum $\vec{u}+\vec{v}$ is in H.
c. H is closed under multiplication by scalars. That is, for each $\vec{u}$ in H and each scalar c, the vector $c\vec{u}$ is in H.

Zero Subspace

The set consisting of only the zero vector in a vector space V is a subspace of V, called the zero subspace and written as $\{\vec{0}\}$.

Linear Combination and Span

- Linear combination refers to any sum of scalar multiples of vectors.
- Span: $\text{Span}\{\vec{v}_1,\vec{v}_2,\dots,\vec{v}_p\}$ denotes the set of all vectors that can be written as linear combinations of $\vec{v}_1,\vec{v}_2,\dots,\vec{v}_p$.

Theorem 1

If $\overrightarrow{v_1}\dots\overrightarrow{v_p}$ are in a vector space V, $\text{Span}\{\vec{v}_1,\vec{v}_2,\dots,\vec{v}_p\}$ is a subspace of V.

We call $\text{Span}\{\vec{v}_1,\vec{v}_2,\dots,\vec{v}_p\}$ the subspace spanned (or generated) by $\{\vec{v}_1,\vec{v}_2,\dots,\vec{v}_p\}$. Given any subspace H of V, a spanning (or generating) set for H is a set $\{\vec{v}_1,\vec{v}_2,\dots,\vec{v}_p\}$ in H such that:
$H=\text{Span}\{\vec{v}_1,\vec{v}_2,\dots,\vec{v}_p\}$

4.2 Null Spaces, Column Spaces, and Linear Transformations

Definition: Null Space

The null space of an $m \times n$ matrix A, written as $\text{Nul } A$, is the set of all solutions of the homogeneous equation $A\vec{x} = \vec{0}$. In set notation:

$$\text{Nul } A = \{\vec{x} \in \mathbb{R}^n | A\vec{x} = \vec{0}\}$$

Theorem 2

The null space of an $m \times n$ matrix is a subspace of $\mathbb{R}^n$.

Equivalently, the set of all solutions to a system $A\vec{x} = \vec{0}$ of m homogeneous linear equations in n unknowns is a subspace of $\mathbb{R}^n$.

Definition: Column Space

The column space of an $m \times n$ matrix A, written as $\text{Col } A$, is the set of all linear combinations of the columns of A. If $A = [\vec{a}_1 \ \cdots \ \vec{a}_n]$, then:

$$\text{Col } A = \text{Span}\{\vec{a}_1, \vec{a}_2, \ldots, \vec{a}_n\}$$

Theorem 3

The column space of an $m \times n$ matrix A is a subspace of R^m

And:

$$\text{Col } A = \{\vec{b} | \vec{b} = A\vec{x} \text{ for some } \vec{x} \in R^n\}$$

The column space of an $m \times n$ matrix A is all of $\mathbb{R}^m$ if and only if the equation $A\vec{x} = \vec{b}$ has a solution for each $\vec{b}$ in $\mathbb{R}^m$.

Definition: Linear Transformation

A linear transformation T from a vector space V into a vector space W is a rule that assigns to each vector $\vec{u}$ in V a unique vector $T(\vec{u})$ in W, such that:

1. $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$ for all $\vec{u}, \vec{v}$ in V
2. $T(c\vec{u}) = cT(\vec{u})$ for all $\vec{u}$ in V and all scalars c

Kernel and Image of a Linear Transformation

- Kernel (or null space) of T: The set of all $\vec{u}$ in V such that:
  $T(\vec{u}) = \vec{0}$
- Image (or range) of T: The set of all vectors $\vec{w}$ in W such that $\vec{w} = T(\vec{u})$ for some $\vec{u}$ in V

Linear Transformation $T(\vec{x}) = A\vec{x}$

If $T(\vec{x}) = A\vec{x}$, then:

- Kernel of T is $\text{Nul } A$
- Image of T is $\text{Col } A$

4.3 Linear Independent Sets, Bases

Theorem 4

A set $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_p\}$ of two or more vectors, with $\vec{v}_1 \neq \vec{0}$, is linearly dependent if and only if some $\vec{v}_j$ (with $j > 1$) is a linear combination of the preceding vectors $\vec{v}_1, \dots, \vec{v}_{j-1}$.

Definition: Basis

Let H be a subspace of a vector space V. An indexed set of vectors $B = \{\vec{b}_1, \dots, \vec{b}_p\}$ in V is a basis for H if:

1. B is a linearly independent set.
2. The subspace spanned by B coincides with H, that is

$$H = \mathrm{Span}\{\vec{b}_1, \dots, \vec{b}_p\}$$

Theorem 5: The Spanning Set Theorem

Let $S = \{\vec{v}_1, \dots, \vec{v}_p\}$ be a set in V, and let $H = \mathrm{Span}\{\vec{v}_1, \dots, \vec{v}_p\}$
a. If one of the vectors in S—say, $\vec{v}_k$—is a linear combination of the remaining vectors in S, then the set formed from S by removing $\vec{v}_k$ still spans H.
b. If $H \neq \{\vec{0}\}$, some subset of S is a basis for H.

Theorem 6

The pivot columns of a matrix A form a basis for $\mathrm{Col}\ A$.

4.4 Coordinate Systems

Theorem 7: The Unique Presentation Theorem

Let $B = \{\vec{b}_1, \dots, \vec{b}_n\}$ be a basis for a vector space V. Then for each $\vec{x}$ in V, there exists a unique set of scalars $c_1, c_2, \dots, c_n$ such that:

$$\vec{x} = c_1\vec{b}_1 + c_2\vec{b}_2 + \cdots + c_n\vec{b}_n$$

Definition: Coordinates Relative to a Basis

Suppose $B = \{\vec{b}_1, \dots, \vec{b}_n\}$ is a basis for V and $\vec{x}$ is in V. The coordinates of $\vec{x}$ relative to the basis B (or the B-coordinates of $\vec{x}$) are the weights $c_1, \dots, c_n$ such that:

$$\vec{x} = c_1\vec{b}_1 + \cdots + c_n\vec{b}_n$$

The vector

$$[\vec{x}]_B = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

is the coordinate vector of $\vec{x}$ (relative to B), or the B-coordinate vector of $\vec{x}$.

The mapping $\vec{x} \mapsto [\vec{x}]_B$ is the coordinate mapping determined by B.

Coordinate Mapping and Change-of-Coordinate Matrix

- Let $B = \{\vec{b}_1, \dots, \vec{b}_n\}$, define $P_B = [\vec{b}_1 \ \cdots \ \vec{b}_n]$. Then the vector equation
  $\vec{x} = c_1\vec{b}_1 + c_2\vec{b}_2 + \cdots + c_n\vec{b}_n$
  is equivalent to:
  $\vec{x} = P_B[\vec{x}]_B$

- P_B is the change-of-coordinate matrix from B to the standard basis in $\mathbb{R}^n$.

Theorem 8

Let $B = \{\vec{b}_1, \ldots, \vec{b}_n\}$ be a basis for a vector space V. Then the coordinate mapping $\vec{x} \mapsto [\vec{x}]_B$ is a **one-to-one** linear transformation from V **onto** $\mathbb{R}^n$.

Isomorphism

In general, a one-to-one linear transformation from a vector space V onto a vector space W is an isomorphism from V onto W.

4.5 The Dimension of a Vector Space

Theorem 9

If a vector space V has a basis $B = \{\vec{b}_1, \ldots, \vec{b}_n\}$, then any set in V containing more than n vectors must be linearly dependent.

Theorem 10

If a vector space V has a basis of n vectors, then every basis of V must consist of exactly n vectors.

Definition: Dimension

- If V is spanned by a finite set, then V is said to have finite dimension, and the dimension of V, written as $\dim V$, is the number of vectors in a basis for V.
- The dimension of the zero subspace $\{\vec{0}\}$ is defined to be 0.
- If V is not spanned by a finite set, then V is said to be infinite-dimensional.

Theorem 11

Let H be a subspace of a finite-dimensional vector space V. Any linearly independent set in H can be extended, if necessary, to a basis for H. Also, H is finite-dimensional and:

$$\dim H \leq \dim V$$

Theorem 12: The Basis Theorem

Let V be a p-dimensional vector space ($p \geq 1$).

- Any linearly independent set of exactly p elements in V is automatically a basis for V.
- Any spanning set of exactly p vectors in V is automatically a basis for V.

Dimension of Null Space and Column Space

- The dimension of $\mathrm{Nul}\ A$ is the number of free variables in the equation $A\vec{x} = \vec{0}$.
- The dimension of $\mathrm{Col}\ A$ is the number of pivot columns in A.

4.6 Rank

Definition: Row Space

The set of all linear combinations of the row vectors of A is called the row space of A and is denoted by $\mathrm{Row}\ A$.

Theorem 13

- If two matrices A and B are row equivalent, then their row spaces are the same.
- If B is in echelon form, the nonzero rows of B form a basis for the row space of A as well as for that of B.

Definition: Rank (1st Definition)

The rank of A is the dimension of the column space of A.

Theorem 14: The Rank Theorem

The dimensions of the column space and the row space of an $m \times n$ matrix A are equal.

The common dimension, the rank of A, also equals the number of pivot positions in A and satisfies the equation:

$$\text{rank } A + \dim \text{Nul } A = n$$

Theorem: The Invertible Matrix Theorem (continued)

Let A be an $n \times n$ matrix. Then the following are equivalent to A being invertible:
m. The columns of A form a basis of $\mathbb{R}^n$
n. $\text{Col } A = \mathbb{R}^n$
o. $\dim \text{Col } A = n$
p. $\text{rank } A = 0$

q. $\text{Nul } A = \vec{0}$

r.$\dim \text{ Nul } A = 0$

Definition: k-order Minor

从矩阵A中任取k行k列，位于这些行列交叉处的k^2个元素，不改变它们在A中的位置而得到的k阶行列式，即为矩阵A的k阶子式。

Definition: Rank (2nd Definition)

设在矩阵A中有一个不为0的r阶子式D，且所有$r+1$阶子式（如存在）全为0，则D为矩阵A的最高非零子式，数r为矩阵A的秩，记作$R(A)$，并规定零矩阵的秩为0。

The Properties of Rank

1. $0 \le R(A_{m \times n}) \le \min\{m, n\}$
2. $R(A^T) = R(A)$
3. **若**A**~**B**，则**$R(A) = R(B)$
4. **若**P**、**Q**可逆，则：**$R(PAQ) = R(A)$

Proof Idea for (3)(4)

- 初等行变换→不改变行秩/列秩；
- 初等列变换（初等对换施行变换）不改变行秩/列秩；
- 若只施行行变换，则P可逆；
- 若只施行列变换，则Q可逆，
- 故：
  $R(PAQ) = R(A)$

5. $\max\{R(A),R(B)\}\leq R(A,B)\leq R(A)+R(B)$

Proof for (5)

由于A的最高阶非零子式为A的非零子式，故$R(A)\leq R(A,B)$，同理$R(B)\leq R(A,B)$。综合，即有：

$$\max\{R(A),R(B)\}\leq R(A,B)$$

设$R(A)=r$，$R(B)=s$，作初等变换使$A\sim A_0$，$B\sim B_0$，则A_0中有r个非零列，B_0中有s个非零列，且$(A,B)\sim(A_0,B_0)$。由于(A_0,B_0)仅有$r+s$个非零列，故：

$$R(A_0,B_0)\leq r+s$$

又由于$R(A,B)=R(A_0,B_0)$，即$R(A,B)\leq r+s$，故：
$R(A,B)\leq R(A)+R(B)$

6. $R(A+B)\leq R(A)+R(B)$

Proof for (6)

$$R(A+B)\leq R\left(\begin{bmatrix}A+B\\B\end{bmatrix}\right)=R\left(\begin{bmatrix}A\\B\end{bmatrix}\right)=R\left(\begin{bmatrix}A^T\cdot B^T\end{bmatrix}^T\right)=R(A^T,B^T)\leq R(A^T)+R(B^T)=R(A)+R(B)$$

7. $R(AB)\leq\min\{R(A),R(B)\}$

Proof for (7)

设A列向量为$\vec{\alpha}_1,\vec{\alpha}_2,\ldots,\vec{\alpha}_n$
$AB=[\vec{\alpha}_1,\vec{\alpha}_2,\ldots,\vec{\alpha}_n]\begin{bmatrix}b_{11}&b_{12}&\ldots&b_{1m}\\b_{21}&b_{22}&\ldots&b_{2m}\\\vdots&\vdots&&\vdots\\b_{n1}&b_{n2}&\ldots&b_{nm}\end{bmatrix}$
$=(\sum_{i=1}^n b_{i1}\vec{\alpha}_i,\sum_{i=1}^n b_{i2}\vec{\alpha}_i,\ldots,\sum_{i=1}^n b_{im}\vec{\alpha}_i)$
故AB列向量可表为A的列向量线性组合，因而
$R(AB)\leq R(A)$，进而有

$$R(AB)=R\left((AB)^T\right)=R\left(B^TA^T\right)\leq R(B^T)=R(B)$$

综上

$$R(AB)\leq\min\{R(A),R(B)\}$$

8. **若**$A_{m\times n}B_{n\times l}=0_{m\times l}$**，则：**$R(A)+R(B)\leq n$

Proof for (8)

设$B=[\vec{b}_1,\vec{b}_2,\ldots,\vec{b}_l]$，则：

$$A\vec{b}_i=\vec{0}\quad(i=1,2,\ldots,l)$$

即$\vec{b}_1,\vec{b}_2,\ldots,\vec{b}_l$都在$\mathrm{Nul}\ A$中。故：

$$R(B)=R(\vec{b}_1,\vec{b}_2,\ldots,\vec{b}_l)\leq\dim\mathrm{Nul}\ A$$

由秩定理$\dim\mathrm{Nul}\ A=n-R(A)$，故：

$$R(A) + R(B) \leq n$$

4.7 Change of Basis

Theorem 15

Let $B = \{\vec{b}_1, \dots, \vec{b}_n\}$ and $C = \{\vec{c}_1, \dots, \vec{c}_n\}$ be bases of a vector space V. Then there exists a unique $n \times n$ invertible matrix $P_{C \leftarrow B}$ such that:

$$[\vec{x}]_C = P_{C \leftarrow B}[\vec{x}]_B$$

for all $\vec{x}$ in V.

Definition of $P_{C \leftarrow B}$

The columns of $P_{C \leftarrow B}$ are the C-coordinate vectors of the vectors in the basis B. That is:

$$P_{C \leftarrow B} = [[\vec{b}_1]_C \ [\vec{b}_2]_C \ \cdots \ [\vec{b}_n]_C]$$

How to Solve $[\vec{b}_j]_C$?

Answer: Solve:

$$[C \ \ \vec{b}_j] \sim [I \ \ [\vec{b}_j]_C]$$

How to Solve $P_{C \leftarrow B}$

For a $P_{C \leftarrow B}$:

$$[C|B] \rightarrow [I|P_{C \rightarrow B}]$$

if C and B are square

Properties of $P_{C \leftarrow B}$

- $P_{C \leftarrow B}$ is the change-of-coordinate matrix from B to C.
- Since $P_{C \leftarrow B}$ is invertible:
  $P_{B \leftarrow C} = (P_{C \leftarrow B})^{-1}$
- For the same vector $\vec{x}$, we have:
  $[\vec{x}]_B = P_{B \leftarrow C}[\vec{x}]_C$

Change of Basis

对于基B_1, B_2，有变化矩阵$P_{1 \leftarrow 2}$，$P_{1 \leftarrow 2}$分别为B_2在B_1下的坐标表示

对任一$\vec{x}$，有 $[\vec{x}]_1 = P_{1 \leftarrow 2}[\vec{x}]_2$

对基组成矩阵，有 $B_2 = B_1 P_{1 \leftarrow 2}$

对同一线性变化，设在B_1下为A_1，在B_2下为A_2
$A_2 = P^{-1} A_1 P$

4.8 Applications to Difference Equations

Definition: Linear Difference Equation

Given scalars $a_0, a_1, \ldots, a_{n-1}$ (with $a_0 \neq 0$) and a sequence $\{z_k\}$, the equation:

$$a_0 y_{k+n} + a_1 y_{k+n-1} + \cdots + a_{n-1} y_{k+1} + a_n y_k = z_k \text{ for all } k$$

is a linear difference equation (or linear recurrence relation) of order n.

Homogeneous and Nonhomogeneous Difference Equations

- If $z_k = 0$ for all k, the equation is homogeneous; otherwise, the equation is nonhomogeneous.

Auxiliary Equation

In general, a nonzero signal r^k satisfies the homogeneous different equation:

$$y_{k+n} + a_1 y_{k+n-1} + \cdots + a_{n-1} y_{k+1} + a_n y_k = 0 \text{ for all } k$$

If and only if r is a root of auxiliary equation.

$$r^n + a_1 r^{n-1} + \cdots + a_{n-1} r + a_n = 0$$

Theorem 16

If $a_n \neq 0$, and if $\{z_k\}$ is given, the equation

$$y_{k+n} + a_1 y_{k+n-1} + a_2 y_{k+n-2} + \cdots + a_{n-1} y_{k+1} + a_n y_k = z_k \text{ for all } k$$

has unique solution whenever $y_0, \ldots, y_{n-1}$ are specified.

Theorem 17.

The set H of all solution of the nth-order homogeneous linear difference equation

$$y_{k+n} + a_1 y_{k+n-1} + \cdots + a_{n-1} y_{k+1} + a_n y_k = 0 \text{ for all } k$$

is an n-dimensional vector space

4.9 Applications to Markov Chains

Basic Definitions

- Probability vector: A vector with nonnegative entries that add up to 1.
- Stochastic matrix: A square matrix whose columns are probability vectors.
- Markov chain: A sequence of probability vectors $\vec{x}_0, \vec{x}_1, \vec{x}_2, \ldots$ together with a stochastic matrix P, such that:
  $\vec{x}_{k+1} = P\vec{x}_k \quad \text{for } k = 0, 1, 2, \ldots$
- State vector: $\vec{x}_k$ is called the state vector at time k.

Steady-State Vector

If P is a stochastic matrix, then a steady-state vector (or equilibrium vector) for P is a probability vector $\vec{q}$ such that:
$P\vec{q} = \vec{q}$

Regular Stochastic Matrix

We say that a stochastic matrix is regular if some power P^k of P contains only strictly positive entries.

Convergence of Vector Sequences

We say that a sequence of vectors $\vec{x}_1, \vec{x}_2, \ldots$ converges to a vector $\vec{q}$ as $k \to \infty$ if the entries in $\vec{x}_k$ can be made as close as desired to the corresponding entries in $\vec{q}$ by taking k sufficiently large.

Theorem 18

If P is an $n \times n$ regular stochastic matrix, then P has a unique steady-state vector $\vec{q}$. Further, if $\vec{x}_0$ is any initial state vector, then:

$$\vec{x}_k = P^k \vec{x}_0$$

for $k = 0, 1, 2, \ldots$, and the Markov chain $\{\vec{x}_k\}$ converges to $\vec{q}$ as $k \to \infty$.